Fulton Hogan Assignment

Software Modelling Report ENSE706

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1. Problem Statement

Fulton Hogan is a New Zealand based construction company that provides quality infrastructure. They are known for constructing and maintaining roads, bridges, energy and water networks all across New Zealand and Australia. Their goals outside of construction have always been “creating, connecting, and caring for communities”.

As of now, Fulton Hogan’s Project Costing & Management section uses JD Edwards to manage their main operations. JD Edwards is an Enterprise Resource Planning system which manages construction projects, equipment, and payroll (Oracle, 2024). The users of this system use it to perform crucial business operations that are important for the successful running of the business. However, it is an old legacy system that is not meant to handle real-time demands of today’s construction management needs.

Each department uses JD Edwards in different ways, however it cannot keep up with modern business-related demands. Fulton Hogan is looking to improve their Project Costing & Management System, which involves three departments; Project Management department, Finance department, and Operations department.

1.1 Project Management Department

The main responsibilities of this department are:

- Creating and managing projects
- Logs timesheets, milestones, and project progress
- Submits budget requests and cost updates

The Project Management department serves as the primary driver of project planning and execution. They are responsible for the entire process of a construction project from start to finish. This department ensures projects are on schedule, and that any tasks are completed within the time given. However, all users of this system have reported latency issues. Whenever Project Coordinators contribute to the project, Project Managers cannot see real-time progress meaning the Site Leads don’t know what to do in current time. This is due to the outdated legacy program, causing scheduling conflicts and data inconsistences.

The main users of this department are Project Managers, Site Leads and Project Coordinators. Project Managers are the people that manage the entire project. When given a project, they are responsible for managing both people and resources. Site

Leads are the people who are on the construction sites and oversee everything going on, making sure everything is following the correct procedure. Project Coordinators are the people working on the project and contributing their input. They are always finding solutions to any problems that arise, making sure everything is working smoothly, and acting as a central point of communication.

The Project Management Department uses the system as the central hub for data entry and project tracking. They initialise projects, log any progress made, and any updates are also triggered by these three user roles. Each user has different privileges which allow them to access different parts of the system. Project Managers can initialise projects and add any major changes to them (extending deadlines, generating team reports, managing the team). However, they cannot see what the Site Leads are telling the Site Foremen. Site Leads use the system to send instructions to the Site Foremen on duty. They need all information available to them and the tasks that are needed to achieve in a selected time period. Project Coordinators use the system to document any new findings, managing new information, analysing data, and creating reports for the Project Manager.

1.2 Finance Department

The main responsibilities of this department are:

- Approving and monitoring project budgets
- Tracking actual vs estimated costs
- Processes financial reporting on projects

The Finance department is responsible for overseeing the financial aspects of project delivery. They ensure that the company is always making profits by reviewing every dollar allocated to the project. This department is always focused on financial data and making sure everything on the financial side is going smoothly. Auditors are always complaining about missing data every now and then. The data was entered, but the auditors could not see the data. This means the data loss was a system error. Fulton Hogan wants this problem to be sorted as a high priority. This is because the data is read by people who make important decisions that lead the project progress forwards.

Financial Controllers, Project Accountants, and Auditors are users for this department. Financial controllers manage the day-to-day financial operations such as monthly reporting, managing cash flow, and ensuring accounting standards are met across project timelines. Project Accountants have the main responsibility of handling financial tracking for projects. Auditors are always reviewing financial records and processes to verify financial accuracy when it comes to project resources.

The Finance Department uses the system as the primary platform for financial analysis and reporting, making sure all expenses remain aligned with approved budgets. They are always comparing Estimated vs Actual (EVA) costs, approving the funds for Project Management requests, and generating monthly compliance reports. Similar to the Project Management Department, the users in this department have different privileges on the system. The Financial Controllers have access to all resources so that they can make profitable decisions. Project Accountants have access to tracking software, however not to the reports that show project spendings. Auditors have access to all financial reports, because they need to have all financial information to create reports for the Project Managers.

1.3 Operations Department

The main responsibilities of this department are:

- Executing the on-ground construction work
- Reporting labour hours and equipment usage
- Flagging project issues that affect cost or timeline

The Operations department is responsible for handling all the physical execution, heavy machinery, labour and materials. Their responsibility is to safely and efficiently deliver the infrastructure for their clients. All users of JD Edwards in this department consistently report issues with system performance. Site Foremen are always having to wait long periods of time for instructions to come through. This slows down progress on the site, since no one can go forward without instructions. Workers have experienced incorrect incomes because their hours weren't logged on properly. Fulton Hogan wants to find the system part responsible for this and remake it.

The main users of this department are the Site Foremen, Heavy Equipment Operators, and General Laborers. The Site Foremen are the supervisors of this department, always managing the day-to-day activities of all the workers. They are constantly translating instructions given by the Site Leads, and other higher ups. Heavy Equipment Operators are in charge of operating all the tools, vehicles, and equipment used to deliver the infrastructure. They use the system for communicating with the Site Foremen and General Laborers, and report anything to do with the equipment and machines. However they should also have access to the instructions given by the Site Lead. This will ensure they are prepared for the upcoming tasks that day and can plan ahead. The General Laborers are the entry level workers who carry out physical tasks on site. They use the system for communication, reporting any problems, and logging in their hours.

This department uses the system as a daily activity log, where everything that happened in the work day is reported and given to people back at the company. The data reported by this department include how many people were working, state of the machines, work to be done the next day, etc. They also have a channel dedicated to reporting issues e.g. broken vehicle or machine.

2. System Overview

Fulton Hogan wants to develop a completely new and updated Project Costing & Management System that allows all three departments to function effortlessly and efficiently. The system should allow any data inputted to go through successfully and without issue, at the same time the other devices using this software should be updated instantaneously. All users should have the correct amount of access, while some users get more privileges than others depending on their role in the company. Others should have updated privileges from the old system, and have more access to important documents that will help them work more effectively. The new Project Costing & Management system should improve efficiency, reduce the time it takes for others to be notified of changes to any data or projects, and provide a much better experience to all system users. This will be done by applying modern software principles that can meet the demands of modern project management.

Below is the table describing all stakeholders of the system.

Stakeholder	Roles	Department
Project Manager	Initialise projects, Manage projects, Generate team reports, Manage the project management team, Receive financial reports from Finance department	Management
Project Coordinator	Log new findings, Manage new information, Analyse data, Create progress reports	Management
Site Lead	Guide Site Foremen during work, Have access to day-to-day instructions, Communicate with project management team	Management
Financial Controller	Access to all financial resources, Make final decisions	Finance

Project Accountant	Access to financial tracking software, Prepare financial statements	Finance
Auditor	Access to all financial reports, issue any financial problems, verify financial accuracy, generate compliance reports	Finance
Site Foremen	Manage Instructions from Site Lead to general labourers, flag any issues	Operations
Heavy Machine Operator	Operate machines, communicate with Site Foremen, Log in hours into timesheet, Access to instructions from Site Lead	Operations
General Labourer	Physical activities, Log in hours into timesheet, flag any issues to the department	Operations

2.1 System Scope

The new Project Costing & Management System should improve real-time data entry, tracking, and reporting across all 3 departments at Fulton Hogan: Project Management, Finance, Operations. Everything mentioned about the departments above will be key features in the scope: Project initialisation and management, timesheet logging, budget management, cost tracking, financial reporting, equipment reporting, issue flagging for equipment, and role-based privileges for the different users of the system.

2.2 System Boundaries

This system is limited to the internal operations of Fulton Hogan's Project Costing & Management function. It will not replace or directly integrate with any other external systems that are used by Fulton Hogan or their users. Human Resources, recruitment, and employee reviews are examples of functions that fall out of the scope of this system. The only functions that are being improved are the Project Management department, Finance department, and Operations department. All other business functions fall outside the scope of this system, and are managed by other software. This was confirmed with the client through consultation during the first

meeting. Based on the consultation, it has been determined that all requirements have been captured within the scope. There are no other missing requirements.

2.3 Assumptions

The following assumptions have been made during the analysis of this system.

1. All users of this new system will have access to a computer (personal or company provided), along with a stable internet connection. This is needed because the system has real-time operations that require a strong connection to see the new updated information.
2. Fulton Hogan will provide detailed training to all of their employees (managers, labourers, office workers) to switch over to the new system, once it has passed testing and has been deployed.
3. Each user will have an account that is registered to them from JD Edwards. All of their data will go through to the new system and be securely stored, so that new accounts do not have to be made. On top of this, there will be a requirement for the user to have a Fulton Hogan specific email, as any other email will be blocked from logging into the new system.
4. Each user role will have their corresponding privileges within the system, these privileges are described in the Problem Statement.
    a. Users in the Project Management department will have exclusive access to tools linked with project management.
    b. Users in the Finance department will have access to various financial tools, allowing them to analyse financial data.
    c. Users in the Operations department will have a timesheet subsystem, along with a flagging issues page, and more.
5. Existing data stored in JD Edwards will be migrated to the new system without any significant losses. It should be noted that Fulton Hogan is committed to maintaining data integrity during the transition period.

3. Requirements Engineering

This section defines both functional and non-functional requirements of the new Project Costing & Management System for Fulton Hogan. Functional requirements describe what the system must do, these are important for the functionality of the system. Non-functional requirements describe the quality standards the system must meet. This could be performance related like update speeds, or it could be related to security and reliability.

3.1 Requirements Elicitation (Interviews and Observations)

Stakeholder: Project Manager

Q: **What issues do you face with the current system?**
A: The current system was good before, but nowadays, it is very slow. We have to usually wait a minute or two before the new info comes in from our team, to analyse it.

Q: **How do these issues affect your work?**
A: It stops us from moving forward you know? Having to wait for our subordinates' work is a pain, because we can't do anything without it. When the system does this, we all get annoyed.

Q: **If you were to get a new system to use, what changes would you want on it?**
A: Definitely good performance, the biggest issue with JD Edwards is the speed. It takes a while to see updated information on projects. The new system should show these changes very fast, maybe 10 seconds max.

Stakeholder: Project Coordinator

Q: **What issues do you face with the current system?**
A: So there are actually quite a few issues right now. When I update my work and research onto the system, it takes a while to upload which is quite a pain. I reckon it shouldn't take this long, 10 seconds max would be nice.

Q: **How often do you upload your work onto the system?**
A: We are always uploading stuff whenever we are working, so it could be 2 to several times a day. Which is why it would be nice to have faster uploads.

Q: **If you were to get a new and updated system for you to use at work, what changes would you have?**
A: Definitely a solution to the upload speeds. While this isn't a problem, I reckon it would be nice to have an updated UI. Our one is very old, and feels like we went back two decades in technology.

Stakeholder: Project Accountant

Q: **What issues do you face with the current system?**

A: The system is very unreliable when it comes to tracking actual versus estimated costs. Me and the rest of my team enter financial data throughout the day, but the Auditors come back to us saying they cannot see some of the entries. From our end, everything was done correctly, but the system just causes problems.

Q: **How does this affect your day-to-day work?**
A: It's really frustrating. Imagine you are just doing your work, and you have to submit something to your manager. Then they say “Where is the work you were meant to submit by 5pm?”. Yeah, it just slows everything down and causes pain to everyone.

Q: **If you were to get a new system, what would you want improved?**
A: We need our entries to be saved and visible to anyone who needs it. When I log something in, the Auditors should be able to see it straight away. Also an updated GUI would make a nice change in environment for me.

Stakeholder: Auditor

Q: **What issues do you face with the current system?**
A: A big one is missing data. I’m always dealing with missing financial entries in the monthly financial reports for a project. Our accountants did confirm they entered the data, so it’s definitely a system issue not a human one.

Q: **How urgent is this issue for your team?**
A: This is a very high priority. The decisions made from the information we review have a huge impact on our projects. Project Managers and Financial Controllers rely on my reports and when they are not complete, progress is slowed.

Q: **What would you want from the new system?**
A: Every financial transaction must be captured without fail. A proper history of who added what entry would make it easier to track where the entry came from. Also we would want to have access to all financial reports so that we can anaylse all information and make our reports more precise.

Stakeholder: Site Foreman

Q: **What issues do you face with the current system?**
A: Well, most of our problems are usually the waiting times for receiving our instructions. Without the instructions from the Site Lead, we cannot move forward.

Q: **How does this affect the work on site?**

A: Productivity goes down a ton. Having to wait when you are all ready to start a task demotivates all the people out here. Everything here depends on those instructions coming to us when needed.

Q: **What would you want from the new system?**
A: Communication needs to be instantaneous. Receiving instructions when we ask should be consistent. A notification for when new instructions arrive would help us out alot aswell. The system needs to be reliable every day, not just sometimes.

Stakeholder: General Labourers

Q: **What issues do you face with the current system?**
A: My hours have not been logging on properly on the system. I enter them at the end of my shifts like I'm told to, but then the pay I get doesn't match them.

Q: **How do you and your colleagues deal with this?**
A: We usually have to contact the project management team, who then let the financial department know. They usually solve this in around 2 days, but even they are getting sick of it.

Q: **What would you want from a new system?**
A: All I want is for my hours to be logged in the system properly and correctly. I'm out here working in the hot sun, so when I get home, all the money I worked for should be there.

3.2 Functional Requirements

Project Management Department:

Description	Stakeholder
The system shall allow Project Managers to initialise a new project.	Project Manager
The system shall allow Project Managers to modify project details.	Project Manager
The system shall allow Project Managers to manage team members.	Project Manager
The system shall allow Project Managers to view real-time progress updates from all departments	Project Manager

The system shall allow Project Coordinators to log progress, findings, and data entries.	Project Coordinator
The system shall allow Project Coordinators to generate reports of their work.	Project Coordinator
The system shall allow Site Leads to view all project tasks and instructions in real time.	Site Lead
The system shall allow Site Leads to communicate instructions directly to Site Foremen.	Site Lead

Finance Department:

Description	Stakeholder
The system shall allow Financial Controllers to approve all project budgets.	Financial Controller
The system shall allow Financial Controllers to access all financial resources.	Financial Controller
The system shall allow Project Accountants to access financial tracking tools.	Project Accountants
The system shall allow Auditors to access all financial reports for verification.	Auditors
The system shall generate monthly financial reports for the Finance Department	Project Accountants and Financial Controllers
The system shall track and compare estimated versus actual project costs in real time.	Project Accountant

Operations Department:

Description	Stakeholder
The system shall allow Site Foremen to receive and view instructions from Site Leads in real time.	Site Foremen
The system shall allow Site Foremen to relay instructions to General Labourers	Site Foremen
The system shall allow Heavy Equipment	Heavy Equipment Operators

Operators to log equipment usage and condition	
The system shall allow Heavy Equipment Operators to view instructions from Site Leads	Heavy Equipment Operators
The system shall allow General Labourers to log daily working hours.	General Labourers
The system shall allow General Labourers to report on-site issues through the system.	General Labourers
The system shall provide a dedicated issue flagging channel for reporting equipment faults.	Operations Department

3.3 Non-Functional Requirements

Performance:

Description	Stakeholder
The system shall synchronise data across all departments in under 10 seconds.	All Departments
The system shall support a minimum of 2000 concurrent users without the performance degrading.	All Users
The system shall process and display timesheet entries within 10 seconds of submission. This is what the developer thinks is a reasonable time, to ensure correct logging.	Operations Department

Reliability:

Description	Stakeholder
The system shall maintain a minimum uptime of 99.9% during business hours.	All Departments
The system shall prevent data loss during processing and transmission between departments.	Finance Department and Operations Department
The system shall maintain a full audit log of all	Finance Department

data entries and modifications.	
The system shall accurately process and record all submitted labour hours without data loss.	Operations Department

Security:

Description	Stakeholder
The system shall encrypt all data transmitted between users and server to hide information from outsiders.	All Users
The system shall restrict access to system features based on the user's role in the company.	All Users
The system shall only allow users with a verified Fulton Hogan email address to sign into the system.	All Users
The system shall comply with the New Zealand Privacy Act 2020.	All Users

Usability:

Description	Stakeholder
The system shall provide tailored user interfaces for each department displaying the tools required for each of them.	All Departments
The system shall show an error message that is in simple English, when an unpredicted input has been entered.	All Users
The system shall provide a modern interface that reduces training time compared to the legacy JD Edwards system.	All Departments
The system shall allow General Labourers to view a summary of their logged hours for the current pay period.	General Labourers

Maintainability:

Description	Stakeholder
The system shall be designed using modern software principles in a modular structure, so a bugged part of the system doesn't affect other parts.	All Departments
The system shall support data migration from JD Edwards without having to manually enter information again.	All Departments

3.4 Requirements Traceability

Req ID	Requirement Description	Stakeholder	Source
FR-PM-01	The system shall allow Project Managers to initialise a new project.	Project Manager	Problem Statement 1.1
FR-PM-02	The system shall allow Project Managers to modify project details.	Project Manager	Problem Statement 1.1 and Project Manager Interview
FR-PM-03	The system shall allow Project Managers to manage team members.	Project Manager	Problem Statement 1.1
FR-PM-04	The system shall allow Project Managers to view real-time progress updates from all departments	Project Manager	Problem Statement 1.1 and Project Manager Interview
FR-PM-05	The system shall allow Project Coordinators to log progress, findings, and data entries.	Project Coordinator	Problem Statement 1.1 and Project Coordinator Interview
FR-PM-06	The system shall allow Project Coordinators to generate reports of their work.	Project Coordinator	Problem Statement 1.1 and Project Coordinator Interview
FR-PM-07	The system shall allow Site Leads to view all	Site Lead	Problem Statement 1.1

	project tasks and instructions in real time.		
FR-PM-08	The system shall allow Site Leads to communicate instructions directly to Site Foremen.	Site Lead	Problem Statement 1.1
FR-F-01	The system shall allow Financial Controllers to approve all project budgets.	Financial Controller	Problem Statement 1.2
FR-F-02	The system shall allow Financial Controllers to access all financial resources.	Financial Controller	Problem Statement 1.2
FR-F-03	The system shall allow Project Accountants to access financial tracking tools.	Project Accountants	Problem Statement 1.2 and Project Accountant Interview
FR-F-04	The system shall allow Auditors to access all financial reports for verification.	Auditors	Problem Statement 1.2 and Auditors Interview
FR-F-05	The system shall generate monthly financial reports for the Finance Department	Auditors and Financial Controllers	Problem Statement 1.2 and Auditors Interview
FR-F-06	The system shall track and compare estimated versus actual project costs in real time.	Project Accountant	Problem Statement 1.2 and Project Accountant Interview
FR-O-01	The system shall allow Site Foremen to receive and view instructions from Site Leads in real time.	Site Foremen	Problem Statement 1.3 and Site Foreman Interview
FR-O-02	The system shall allow Site Foremen to relay instructions to General Labourers	Site Foremen	Problem Statement 1.3 and Site Foreman interview
FR-O-03	The system shall allow Heavy Equipment Operators to log	Heavy Equipment Operators	Problem Statement 1.3

	equipment usage and condition		
FR-O-04	The system shall allow Heavy Equipment Operators to view instructions from Site Leads	Heavy Equipment Operators	Problem Statement 1.3
FR-O-05	The system shall allow General Labourers to log daily working hours.	General Labourers	Problem Statement 1.3 and General Labourer Interview
FR-O-06	The system shall allow General Labourers to report on-site issues through the system.	General Labourers	Problem Statement 1.3
FR-O-07	The system shall provide a dedicated issue flagging channel for reporting equipment faults.	Operations Department	Problem Statement 1.3
NFR-P-01	The system shall synchronise data across all departments in under 10 seconds.	All Departments	Project Manager Interview and Project Coordinator Interview
NFR-P-02	The system shall support a minimum of 2000 concurrent users without the performance degrading.	All Users	Developer's decision of expected organisational user base
NFR-P-03	The system shall process and display timesheet entries within 10 seconds of submission. This is what the developer thinks is a reasonable time, to ensure correct logging.	Operations Department	Developer's decision, following the project manager's 10 second requirement.
NFR-R-01	The system shall maintain a minimum uptime of 99.9% during business hours.	All Departments	Problem Statement 1.1, 1.2, 1.3 all report issues.
NFR-R-02	The system shall prevent data loss during	Finance Department and	Problem Statement 1.2,

	processing and transmission between departments.	Operations Department	Auditor Interview and General Labourer Interview
NFR-R-03	The system shall maintain a full audit log of all data entries and modifications.	Finance Department	Problem Statement 1.2 and Auditor Interview
NFR-R-04	The system shall accurately process and record all submitted labour hours without data loss.	Operations Department	Problem Statement 1.3 and General Labourer Interview
NFR-S-01	The system shall encrypt all data transmitted between users and server to hide information from outsiders.	All Users	Problem Statement 1.2. Data loss is a high priority
NFR-S-02	The system shall restrict access to system features based on the user's role in the company.	All Users	Problem Statement 1.1, 1.2 and 1.3. All departments have different uses for the system
NFR-S-03	The system shall only allow users with a verified Fulton Hogan email address to sign into the system.	All Users	System Assumption 3
NFR-S-04	The system shall comply with the New Zealand Privacy Act 2020.	All Users	Employment New Zealand (2025)
NFR-U-01	The system shall provide tailored user interfaces for each department displaying the tools required for each of them.	All Departments	Problem Statement 1.1, 1.2, 1.3. Each department uses the system for different tasks.
NFR-U-02	The system shall show an error message that is in simple English, when an unpredicted input has been entered.	All Users	Developer's decision. Good practice for system use
NFR-U-03	The system shall provide a modern interface that is	All Departments	Project Accountant Interview and

	easier to use, and reduces training time compared to the legacy JD Edwards system.		Project Coordinator Interview
NFR-U-04	The system shall allow General Labourers to view a summary of their logged hours for the current pay period.	General Labourers	General Labourer Interview and Problem Statement 1.3
NFR-M-01	The system shall be designed using modern software principles in a modular structure, so a bugged part of the system doesn't affect other parts.	All Departments	Problem Statement. Legacy system failures affect all departments.
NFR-M-02	The system shall support data migration from JD Edwards without having to manually enter information again.	All Departments	System Assumption 8

References

Fulton Hogan. (2024). Retrieved from https://www.fultonhogan.com/

Oracle. (2024). JD Edwards EnterpriseOne, Retrieved from https://www.oracle.com/anz/applications/jd-edwards-enterpriseone/#rc30p4

Employment New Zealand. (2025). Employee Privacy, Retrieved from https://www.employment.govt.nz/fair-work-practices/employee-privacy